

Bipeng Wang

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SUMMARY

Ph.D. in Chemical Engineering with expertise in computational materials science and scientific machine learning. Developed physics-informed machine learning methods to accelerate electronic structure calculations and nonadiabatic molecular dynamics, resulting in seven first-author publications. Experienced in first-principles simulations, high-performance computing, and AI-driven materials research.

EDUCATION

09/2019-08/2025 **Ph.D. (Chemical Engineering)**, *Univ. of Southern California, Los Angeles, USA*

01/2021-12/2023 **M.S. (Computer Science)**, *Univ. of Southern California, Los Angeles, USA*

09/2017-05/2019 **M.S. (Chemical Engineering)**, *Univ. of Southern California, Los Angeles, USA*

09/2013-05/2017 **B.S. (Applied Chemistry)**, *Shenyang Univ. of Chemical Technology, Shenyang, China*

PROJECT EXPERIENCE 2019-present

Physics-informed Machine Learning for Quantum Dynamics

- Developed ANN surrogate models for excited-state Hamiltonian interpolation, reducing expensive ab initio calculations by over two orders of magnitude while maintaining meV-level prediction accuracy.
- Proposed inverse FFT reconstruction and Bi-LSTM sequence learning methods for long-horizon Hamiltonian prediction from sparsely sampled molecular dynamics trajectories.
- Built scalable machine learning pipelines integrating VASP, LAMMPS, DeepMD-kit, Libra, and PyTorch for quantum dynamics simulations on HPC clusters.
- Enabled nanosecond-scale excited-state simulations previously infeasible with direct electronic structure calculations.
- Published 5 first-author papers on machine learning accelerated quantum dynamics.

Defect Physics and Electronic Structure Modeling

- Investigated defect-induced electronic structure fluctuations in metal halide perovskites using long-timescale quantum molecular dynamics simulations.
- Revealed electron-volt-scale fluctuations of defect levels and their impact on charge trapping and carrier recombination.
- Demonstrated sub-bandgap carrier harvesting and energy up-conversion mechanisms from first-principles simulations.
- Combined DFT, molecular dynamics, and excited-state simulations to understand defect-mediated optoelectronic properties.

Machine Learning for Scientific Discovery

- Developed outlier detection algorithms to identify rare events from large-scale quantum molecular dynamics trajectories.
- Designed uncertainty-aware workflows for scientific data analysis and anomaly discovery.
- Built an LLM-powered scientific assistant integrating retrieval-augmented generation (RAG), spectroscopy analysis, semantic search, and automated report generation.
- Developed full-stack AI workflows combining FastAPI, vector databases, and large language models for scientific knowledge discovery.

SKILLS

Scientific Computing: Density Functional Theory, Nonadiabatic Molecular Dynamics, Electronic Structure Calculations, Molecular Dynamics, VASP, CP2K, LAMMPS, DeepMD-kit, Libra, PyXAID, ASE, pymatgen

Machine Learning: PyTorch, Neural Networks, Physics-informed Machine Learning, Scientific Machine Learning, Time-series Modeling, Representation Learning, Uncertainty Quantification, Anomaly Detection

Programming: Python, NumPy, SciPy, Pandas, Scikit-learn, Linux, Git

High-Performance Computing: HPC Clusters, Slurm, Parallel Computing, GPU Computing

Research Areas: Computational Materials Science, Electronic Structure Theory, Quantum Dynamics, Machine Learning for Atomistic Simulation, Defect Physics

PUBLICATIONS

Machine Learning Model Development:

- Wang, B. P. et al. Interpolating Nonadiabatic Molecular Dynamics Hamiltonian with **Artificial Neural Networks**. *J. Phys. Chem. Lett.* **2021**, 12, 6070-6077. ([Link](#))
- Wang, B. P. et al. Interpolating Nonadiabatic Molecular Dynamics Hamiltonian with **Bidirectional Long Short-Term Memory Networks**. *J. Phys. Chem. Lett.* **2023**, 14, 7092-7099. ([Link](#))
- Wang, B. P. et al. Identifying Rare Events in Quantum Molecular Dynamics of Nanomaterials with Outlier Detection Indices. *J. Phys. Chem. Lett.* **2024**, 15, 10384-10391. ([Link](#))

Scalable Computational Frameworks:

- Wang, B. P. et al. Interpolating Nonadiabatic Molecular Dynamics Hamiltonian with **Inverse Fast Fourier Transform**. *J. Phys. Chem. Lett.* **2022**, 13, 331-338. ([Link](#))
- Wang, B. P. et al. Efficient Modeling of Quantum Dynamics of Charge Carriers in Materials Using Short Nonequilibrium Molecular Dynamics. *J. Phys. Chem. Lett.* **2023**, 14, 8289-8295. ([Link](#))
- Wang, B. P. et al. Electron-Volt Fluctuation of Defect Levels in Metal Halide Perovskites on a 100 Ps Time Scale. *J. Phys. Chem. Lett.* **2022**, 13, 5946-5952. ([Link](#))
- Wang, B. P. et al. Sub-Bandgap Charge Harvesting and Energy up-Conversion in Metal Halide Perovskites: Ab Initio Quantum Dynamics. *Npj Comput. Mater.* **2025** ([Link](#))